Runbook: Restoring the Student Records Server from Backup

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1. Purpose and scope

This runbook describes how to restore the student records service after a disk failure or data corruption on the primary server. It applies to the production server only; the test server is rebuilt from scratch instead. Follow the steps in order and record each completed step in the change ticket.

2. Systems involved

Servers that take part in the backup and restore cycle

Host	Role	Address	Backup window
`rec-prod-01`	Student records, primary	`192.0.2.21`	01:30–02:15
`rec-db-01`	Database for records	`192.0.2.22`	01:00–01:30
`bak-nas-02`	Backup storage	`192.0.2.40`	not applicable

3. Before you start

1. Announce the maintenance window to the registry office at least 24 hours in advance.
2. Check that the most recent backup completed without errors in the backup report.
3. Make sure you have console access to `rec-prod-01` in case the network fails.

> **Warning:** Do not restore while the enrolment period is open. A restore replaces every record changed since the last backup, and changes made by registry staff during the day would be lost.

4. Stop the service and mount the backup

Log in to `rec-prod-01` as an administrator and run the following commands. The last command lists the available archives so that you can choose the correct date.

```
sudo systemctl stop recordsd
sudo mount /dev/sdb1 /mnt/backup
ls -lh /mnt/backup/records/
```

Archives are named by date in the form `records-YYYY-MM-DD.tar.gz`. Choose the most recent archive taken before the fault was noticed.

5. Restore the data

Extract the chosen archive over the data directory and give the files back to the service account. Replace the date in the file name with the archive you selected.

```
cd /var/lib/records
sudo tar -xzf /mnt/backup/records/records-2025-11-02.tar.gz --strip-components=1
sudo chown -R records:records /var/lib/records
sudo systemctl start recordsd
```

6. Check the backup configuration

After a restore the backup schedule must still be active. Compare the configuration file with the reference copy below; the line numbers are shown for discussion only.

```
# /etc/recordsd/backup.yaml
schedule: "30 1 * * *"
target: /mnt/backup/records
retention_days: 21
compress: gzip
notify:
  email: it-alerts@eastfield-college.example
  on_failure_only: true
```

> **Note:** The schedule uses cron syntax and runs every night at 01:30. Keeping 21 days of archives needs about 40 GB on the backup storage.

7. Verify and close

Query the health endpoint of the service. A healthy service answers with the status and the database connection state.

```
curl -s http://localhost:8080/health
# expected: {"status":"ok","db":"connected"}
```

1. Ask a member of the registry office to open three recent student files and confirm that they are complete.
2. Unmount the backup disk with `sudo umount /mnt/backup`.
3. Close the change ticket and note the archive date that was restored.

If the health check fails twice, stop and escalate to the on-call infrastructure engineer. Do not attempt a second restore without their approval.